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Undergraduate Vacation Studentships Selection Panel
CSIRO Technology

Re: Application, 2026/27 Undergraduate Vacation Studentship (Technology)
Preference 1: Technology 11, Human-in-the-Loop Planning for Heterogeneous Robot Teams
Preference 2: Technology 14, Learning Engine and Fleet Knowledge from Robot Experience, HART OS

Dear Selection Panel,

When I was younger, I wanted to be a scientist.

I did not really understand what being a scientist meant. To me, it meant finding little experiments on YouTube, gathering whatever I could find around the house, and trying them myself. Sometimes they worked. Sometimes they did not. Either way, I loved changing something, watching what happened, and feeling, however naively, as though I had discovered something for myself.

I eventually stopped telling people I wanted to be a scientist. I do not think I ever grew out of the reason I wanted to be one.

I felt it again in Year 12 while completing my IB Psychology Extended Essay. This was very different from the experiments I had played with as a child. I had one question, months to work on it, and a body of research I had to read, understand and eventually turn into a conclusion I was willing to defend.

I remember becoming completely absorbed in it. I would follow one paper into another, find something that challenged what I thought I understood, go backwards, reconsider my thinking, and slowly try to make the pieces fit together. I received an A for the essay, which I was proud of, but what stayed with me was the experience of spending months thinking deeply about essentially one question and somehow enjoying that process.

There are not many things that make me lose track of time. Being completely invested in solving one problem does.

Years later, I found that feeling again in engineering.

During one trimester at Deakin, I was completing two unrelated final projects. For Backend Development, I was building a server-side platform for controlling robots. For Embedded Systems, I was building a physical two-wheel robot using an Arduino Nano, Raspberry Pi and several sensors.

The projects were never supposed to interact. I had an idea that they could.

When I spoke to my backend professor about connecting them, he was supportive, but also honest with me. What I was proposing was ambitious. Both projects were significant pieces of assessment, and trying to combine them created a real possibility that I would spread myself too thin, burn out, and fail to properly finish either one.

I knew he was right. There was a much safer option in front of me. I could finish each project independently, meet the requirements, take the marks and move on.

But I could already see the combined system in my head, and I needed to know whether I could make it real.

So I took the risk.

For about a month, I built, tested, broke and rebuilt the system. Sometimes the backend worked and the robot ignored it. Sometimes the robot moved while the software believed something completely different had happened. I would solve one problem and discover that I had created another somewhere I thought I had already finished. There were moments when I was frustrated enough to wonder why I had made two important projects unnecessarily difficult.

But I have learned something about myself when I get frustrated by a problem. It becomes very difficult for me to leave it alone.

I change something. I test it. I break something else. I go backwards. I try again.

Eventually, the backend spoke to the robot.

And the robot listened.

The backend project received 100 out of 100 and was selected from my cohort for showcase at Deakin Open Day. I am extremely proud of that result, but what mattered more to me was proving that an idea which had existed only in my head could actually be built.

Looking back, I see a pattern.

The child copying science experiments from YouTube, the Year 12 student buried in psychology literature, and the engineering student refusing to leave a broken robot alone were doing very different things. What I enjoyed underneath them was remarkably similar.

I like finding solutions. I like testing whether an idea is actually true. I like experimenting and seeing what happens when an assumption meets reality. I like building things, breaking them, understanding why they broke, and trying again. Most of all, I like that moment when something I did not understand finally starts to make sense.

That is a part of myself I genuinely like.

It is also the main reason I am applying to CSIRO.

I know research is much more than experimenting until something works. Good research requires rigorous methodology, careful documentation, analysis, reproducibility, and responsible experimental design. These are disciplines I have only begun to experience.

But underneath that structure is something I recognise. There is a question you do not yet know the answer to. You work out how to investigate it. You test what you believe. Sometimes you discover that you were wrong. Then you follow the evidence and try to understand why.

That process feels familiar to me.

I do not want to pretend that I already know research is the career for me. I do not. My Software Engineering (Honours) degree will eventually give me formal exposure to research, but I want to experience it in an environment where research is the actual work and the answer is genuinely unknown. I want to find out whether the feeling I had during my Extended Essay and have since rediscovered through engineering is something I want to build a career around.

That is also why Technology 11 and Technology 14 are my first two preferences. They are the two robotics-based research projects, and they sit close to the kind of work I have already discovered I enjoy. Two career paths I am seriously considering are robotics and backend engineering, and much of my university work has naturally ended up somewhere between the two.

Technology 11 interests me because I have already experienced the gap between controlling a robot in software and watching that plan meet the physical world. Technology 14 interests me because my backend work has made me equally interested in what happens after execution: recording what a robot actually did, reconstructing

trustworthy state, and using that history to understand what happened. I do not expect my university projects to be equivalent to either research problem, but they have given me enough exposure to know that these are questions I genuinely want to spend more time around.

These studentships would therefore let me answer two questions at once. I could experience research properly, while doing so in an area I am seriously considering building my career in.

There will be things I do not know. I have not used ROS or ROS2 before. There will be research methods, robotics concepts and tools that are new to me. That does not bother me. If anything, I know myself well enough now to know what usually happens when I encounter something I do not understand.

I struggle with it. I get frustrated with it. Then I keep working until I understand more than I did when I started.

I cannot tell you today whether I will finish my degree wanting to become a researcher, a robotics engineer, a backend engineer, or something I have not encountered yet.

What I can tell you is that some of the most rewarding experiences I have had, from childhood experiments, to my Extended Essay, to building and breaking robots at university, have involved becoming completely absorbed in a question and refusing to leave it unexplored.

I would like the opportunity to find out where that instinct can take me next.

I am Melbourne-based, able to work on site at Clayton daily, and available full time from November 2026 to February 2027. My project demonstrations and source code are linked in my résumé.

Thank you for considering my application.

Yours sincerely,

Manit Khera